

Introduction to Economic Fluctuations

Chapter 12

Econ 313 – Intermediate Macroeconomics

University of Tennessee, Knoxville

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IN THIS CHAPTER, YOU WILL LEARN:

- ▶ Facts about the business cycle
- ▶ How the short run differs from the long run
- ▶ An introduction to aggregate demand
- ▶ An introduction to aggregate supply in the short run and in the long run
- ▶ How the model of aggregate demand and aggregate supply can be used to analyze the short-run and long-run effects of “shocks.”

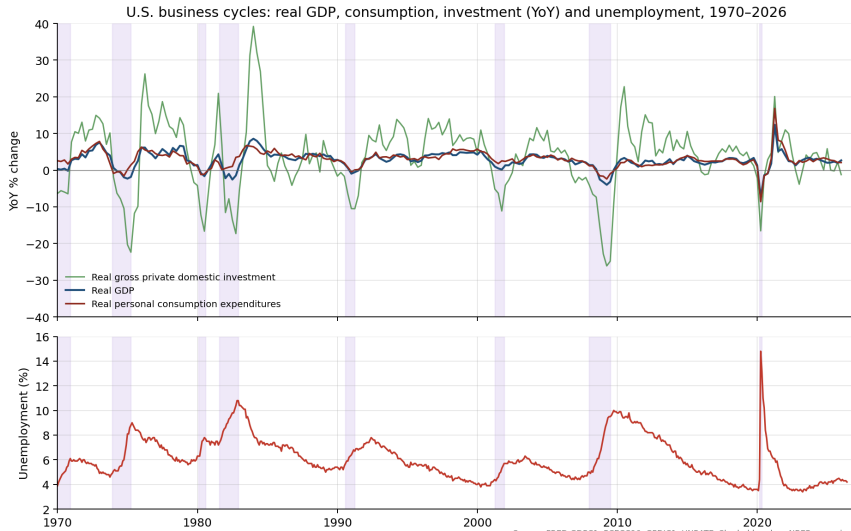
Facts about the business cycle

- ▶ GDP growth averages 3 percent per year over the long run, with large fluctuations in the short run.
- ▶ Consumption and investment fluctuate with GDP, but consumption tends to be less volatile and investment more volatile than GDP.
- ▶ Unemployment rises during recessions and falls during expansions.
- ▶ **Okun's law:** the negative relationship between GDP and unemployment



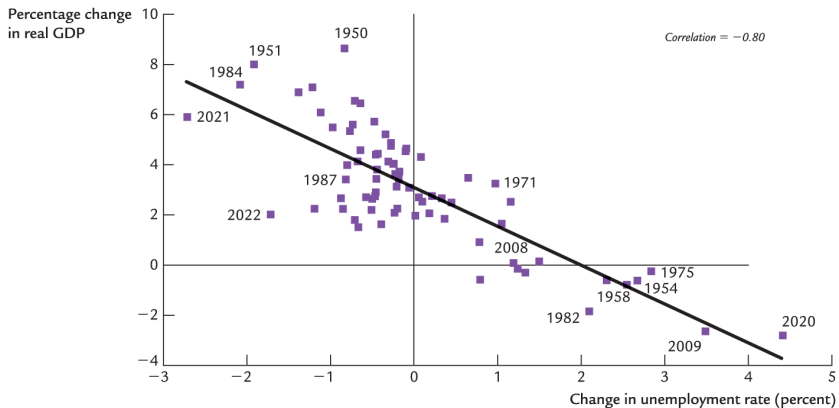
Wall Street outside the NYSE, October 1929. The Great Depression is the textbook business-cycle downturn.

U.S. business cycles: GDP, consumption, investment, unemployment



Sources: FRED GDPIC1, PCECC96, GPDIC1, UNRATE. Bands = NBER recessions.

Okun's law



► Numerical example

Index of leading economic indicators

- ▶ Published monthly by the Conference Board
- ▶ Aims to forecast changes in economic activity six to nine months into the future
- ▶ Used in planning by businesses and government, despite not being a perfect predictor

Components of the LEI (Leading Economic Index)

- ▶ Average weekly hours in manufacturing
- ▶ Average weekly initial claims for unemployment insurance
- ▶ Manufacturers' new orders for consumer goods and materials
- ▶ Manufacturers' new orders, nondefense capital goods, excluding aircraft
- ▶ **ISM** (Institute for Supply Management) new orders index
- ▶ Building permits for new private housing units
- ▶ Index of stock prices
- ▶ Lending credit index
- ▶ Interest rate spread (10-year Treasury – federal funds rate)
- ▶ Average consumer expectations for business and economic conditions

Time horizons in macroeconomics

- ▶ Long run: prices are flexible, responding to changes in supply or demand.
- ▶ Short run: many prices are “sticky” at a predetermined level.
- ▶ *The economy behaves much differently when prices are sticky.*

Recap of classical macro theory (Chapters 3–11)

Output is determined by the supply side:

- ▶ supplies of capital, labor
 - ▶ technology
- ▶ Changes in demand for goods and services (C , I , G) only affect prices, not quantities.
- ▶ Assumes complete price flexibility.
- ▶ Applies to the long run.

When prices are sticky

. . . output and employment also depend on demand, which is affected by:

- ▶ fiscal policy (G and T)
 - ▶ monetary policy (M)
 - ▶ other factors, like exogenous changes in C or I

The model of aggregate demand and supply

- ▶ The paradigm most mainstream economists and policymakers use to think about economic fluctuations and policies to stabilize the economy
- ▶ Shows how the price level and aggregate output are determined
- ▶ Shows how the economy's behavior is different in the short run and in the long run

Aggregate demand

- ▶ The aggregate demand curve shows the relationship between the price level and the quantity of output demanded.
- ▶ For this chapter's intro to the *AD/AS* model, we use a simple theory of aggregate demand based on the quantity theory of money.
- ▶ Chapters 12–14 develop the theory of aggregate demand in more detail.

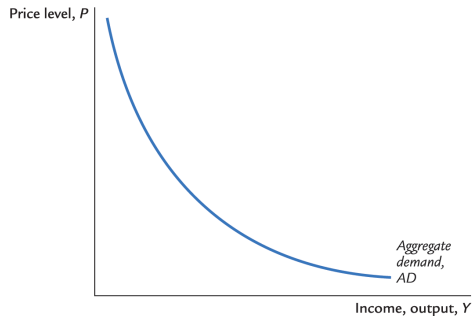
The quantity equation as aggregate demand

From Chapter 5, recall the quantity equation:

- ▶ $MV=PY$
- ▶ For given values of M and V , this equation implies an inverse relationship between P and Y . . .

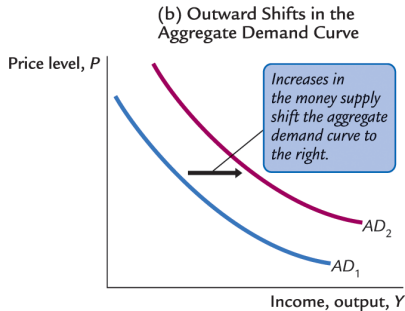
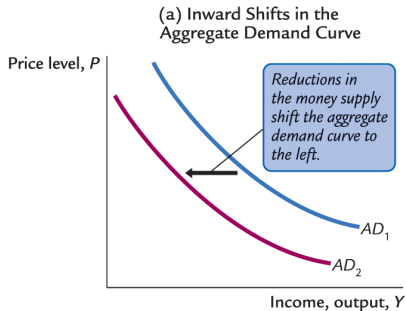
The downward-sloping AD curve

- ▶ An increase in the price level causes a fall in real money balances (M/P), causing a decrease in the demand for goods and services.



Shifting the AD curve

- ▶ A reduction in the money supply shifts the aggregate demand curve to the left.

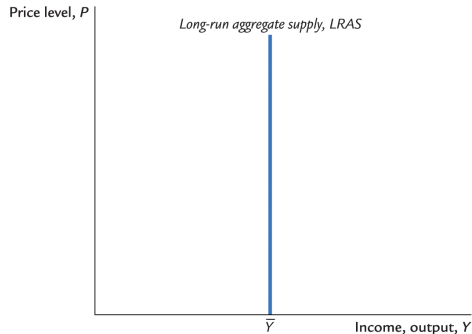


Aggregate supply in the long run

- ▶ Recall from Chapter 3: In the long run, output is determined by factor supplies and technology.
- ▶ is the **full-employment** or **natural** level of output, at which the economy's resources are fully employed.
- ▶ *“Full employment” means that unemployment equals its natural rate (not zero).*

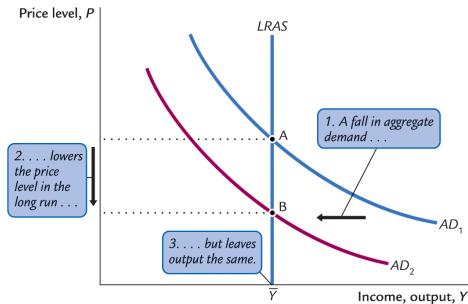
The long-run aggregate supply curve

- ▶ In the long run, output is determined by the amounts of capital and labor and by the available technology; it does not depend on the price level. Therefore, the long-run aggregate supply (LRAS) curve is vertical.



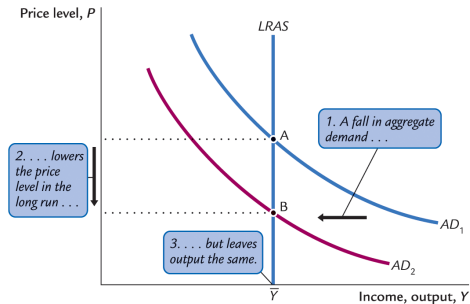
Long-run effects of a decrease in M (1 of 3)

- Starting in the initial equilibrium at A , M down shifts AD in. The economy reaches a new equilibrium B , with a lower price level, and the same level of output.



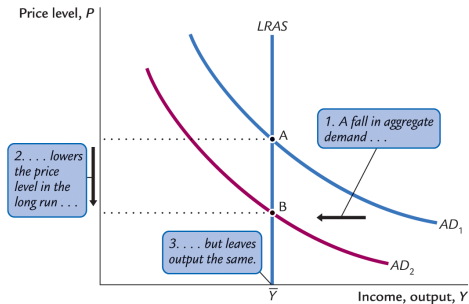
Long-run effects of a decrease in M (2 of 3)

- Then M down, causes AD to shift in



Long-run effects of a decrease in M (3 of 3)

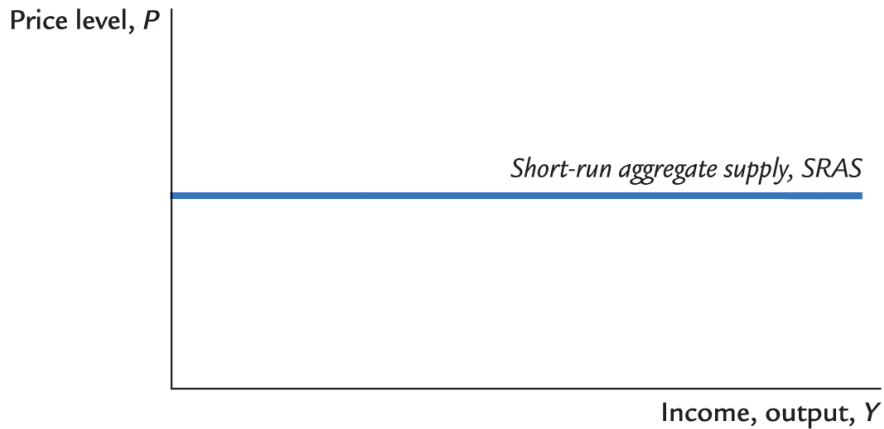
- ▶ With a new equilibrium at B , which has a lower price level, and the same level of output



Aggregate supply in the short run

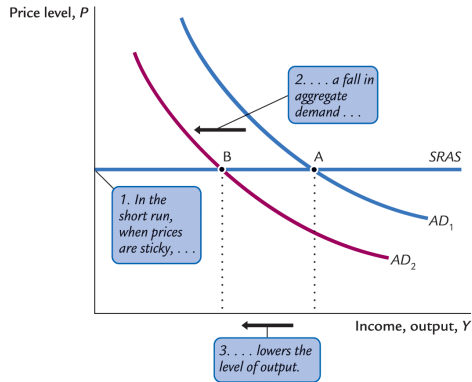
- ▶ Many prices are sticky in the short run.
- ▶ For now (and through Chapter 14), we assume
 - ▶ all prices are stuck at a predetermined level in the short run.
 - ▶ firms are willing to sell as much at that price level as their customers are willing to buy.
- ▶ Therefore, the short-run aggregate supply (*SRAS*) curve is horizontal.

The short-run aggregate supply curve



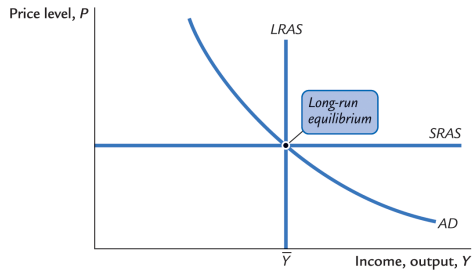
Short-run effects of a decrease in M (1 of 4)

- Starting in initial equilibrium at A , M decreases. M down causes AD to shift in. The inward shift causes the economy to move to a new equilibrium at B , where output falls and price level remains the same.



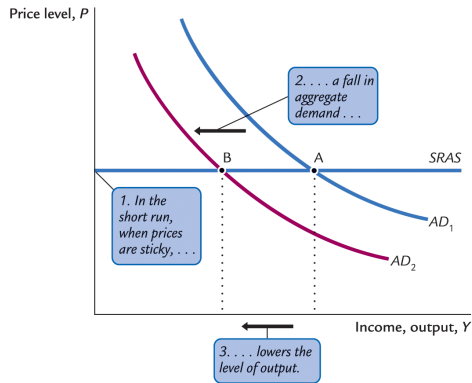
Short-run effects of a decrease in M (2 of 4)

- ▶ Starting in initial equilibrium at A



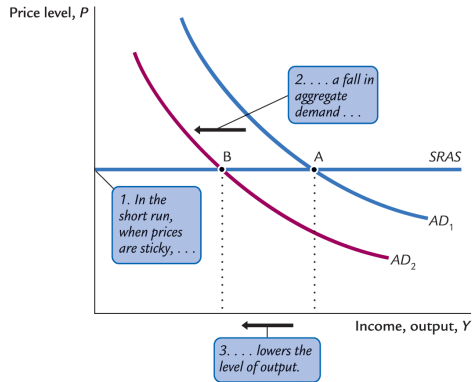
Short-run effects of a decrease in M (3 of 4)

- M down causes AD to shift in.



Short-run effects of a decrease in M (4 of 4)

- The inward shift in AD causes the economy to move to a new equilibrium at B , where output falls and price level remains the same.

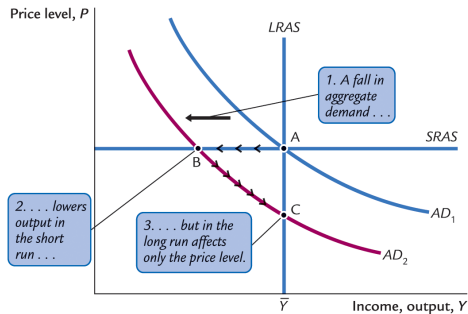


From the short run to the long run

- ▶ Over time, prices gradually become “unstuck.” When they do, will they rise or fall?
- ▶ *The adjustment of prices is what moves the economy to its long – run equilibrium.*

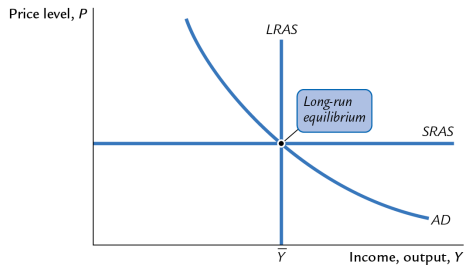
The short- and long-run effects of a decrease in M (1 of 5)

- ▶ The inward shift in AD causes the economy to move to a new short-run equilibrium at B , and then prices adjust and the economy reaches the new long-run equilibrium at C .



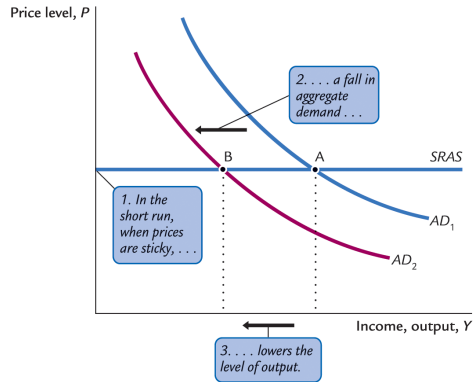
The short- and long-run effects of a decrease in M (2 of 5)

► The initial equilibrium



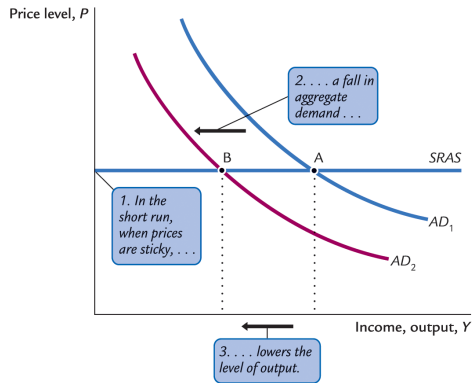
The short- and long-run effects of a decrease in M (3 of 5)

- The decrease in M causes AD to shift in.



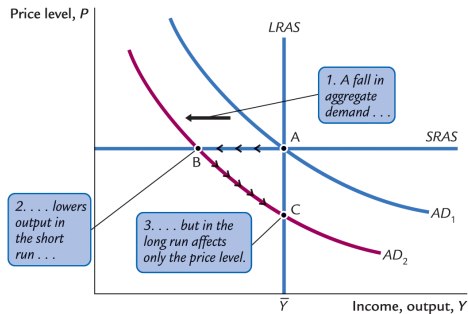
The short- and long-run effects of a decrease in M (4 of 5)

- Causing a new *short-run* equilibrium at B . Price level remains unchanged, but output falls.



The short- and long-run effects of a decrease in M (5 of 5)

- In the long run, prices adjust and the economy reaches a new *long-run* equilibrium at C . Price level is lower, and output returns to

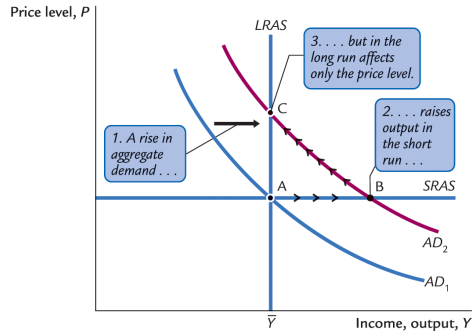


How shocking!

- ▶ **Shocks:** exogenous changes in aggregate supply or demand
- ▶ Shocks temporarily push the economy away from full employment.
- ▶ For example: exogenous decrease in velocity
 - ▶ If the money supply is held constant, a decrease in V means people will be using their money in fewer transactions, causing a decrease in demand for goods and services.

The effects of a positive demand shock

- ▶ A positive demand shock shifts AD out to AD_2 , and the economy has a new short-run equilibrium at B .
- ▶ The economy self-adjusts with increased prices and causes the economy to reach the new long-run equilibrium at C .



Supply shocks

- ▶ A **supply shock** alters production costs and affects the prices that firms charge (also called **price shocks**).
- ▶ Examples of *adverse* supply shocks:
 - ▶ Bad weather reduces crop yields, pushing up food prices.
 - ▶ Oil cartel raises the price of oil.
- ▶ *Favorable* supply shocks lower costs and prices.

CASE STUDY: The 1970s oil shocks, part 1

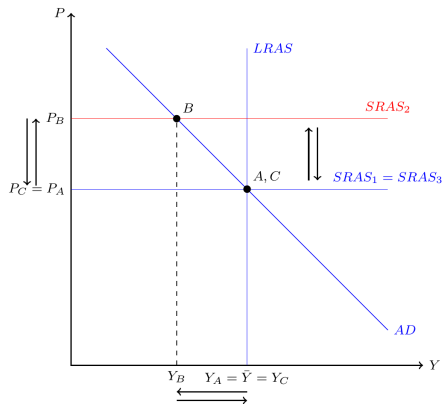
- ▶ Early 1970s: OPEC coordinated a reduction in the supply of oil.
- ▶ Oil prices rose
- ▶ 11 percent in 1973
- ▶ 68 percent in 1974
- ▶ 16 percent in 1975
- ▶ Such sharp oil price increases are supply shocks because they significantly impact production costs and prices.



A driver in Jan 1974 reads about gas rationing in front of a "Sorry, No Gasoline" sign.

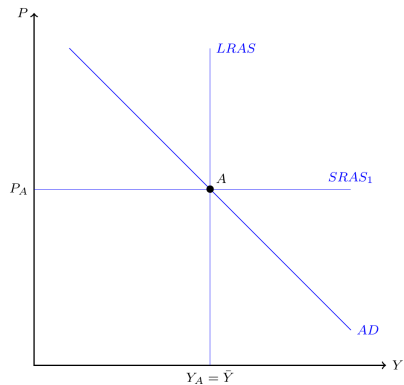
CASE STUDY: The 1970s oil shocks, part 2 (1 of 4)

- ▶ An increase in oil prices causes *SRAS* to shift up. This causes a new short-run equilibrium at *B*.
- ▶ In the long run, prices adjust and cause *SRAS* to shift down back to the original position, reaching a new long-run equilibrium at *C* (which is the same as *A*)



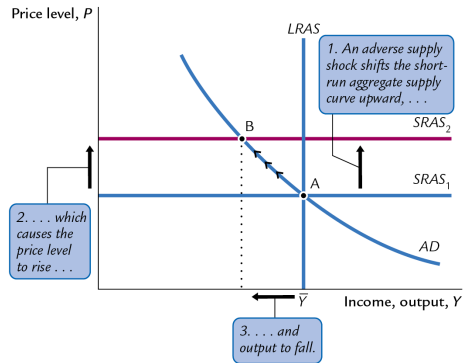
CASE STUDY: The 1970s oil shocks, part 2 (2 of 4)

- The initial equilibrium at A .



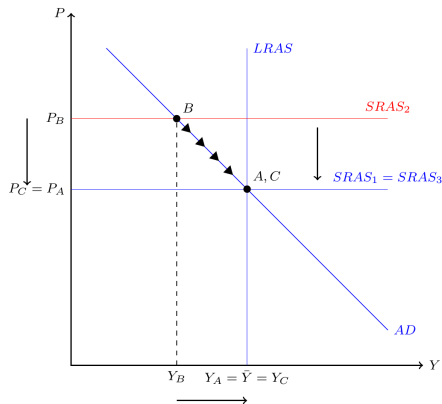
CASE STUDY: The 1970s oil shocks, part 2 (3 of 4)

- ▶ An increase in oil prices causes $SRAS$ to shift up. This causes a new short-run equilibrium at B .
- ▶ Price level increases and output falls in the short run.



CASE STUDY: The 1970s oil shocks, part 2 (4 of 4)

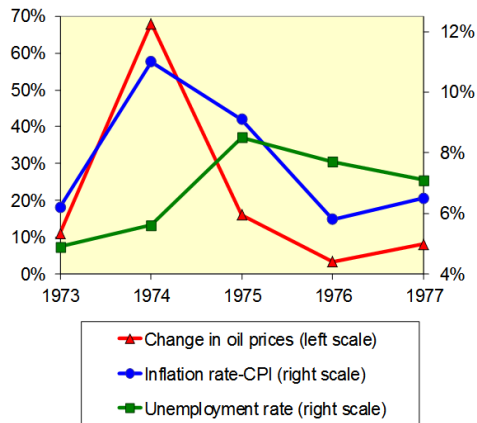
- In the long run, prices adjust and cause *SRAS* to shift down back to the original position, reaching a new long-run equilibrium at *C* (which is the same as *A*).



CASE STUDY: The 1970s oil shocks, part 3

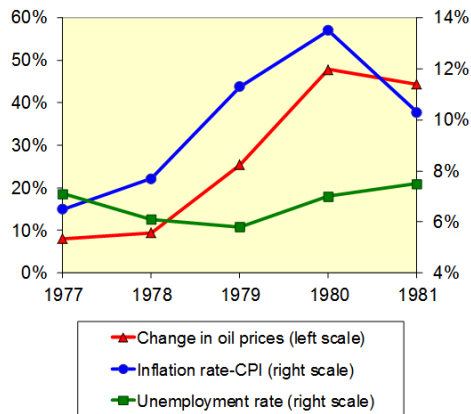
Predicted effects of the oil shock:

- ▶ inflation rate up
 - ▶ output down
 - ▶ unemployment up
- ▶ . . . and then a gradual recovery



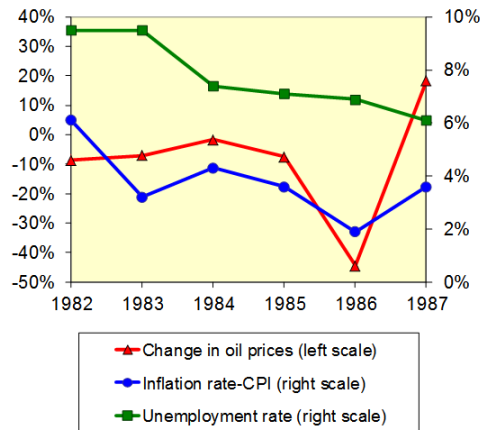
CASE STUDY: The 1970s oil shocks, part 4

- Late 1970s: As the economy was recovering, oil prices shot up again, causing another huge supply shock!



CASE STUDY: The 1980s oil shocks

- ▶ 1980s: A favorable supply shock—a significant fall in oil prices
- ▶ As the model predicts, inflation and unemployment fell.

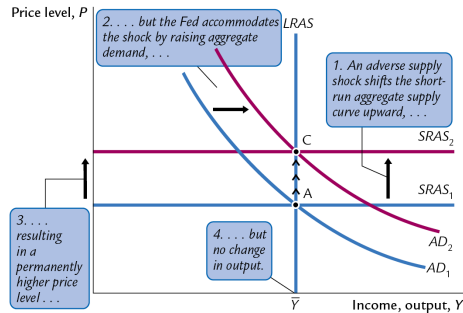


Stabilization policy

- ▶ **Stabilization policy:** policy actions aimed at reducing the severity of short-run economic fluctuations
- ▶ For example: using monetary policy to combat the effects of adverse supply shocks

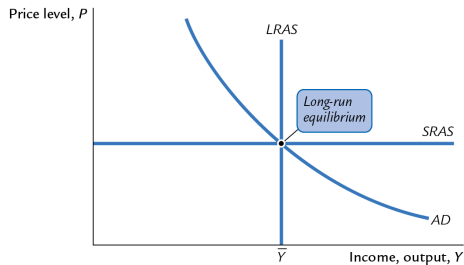
Stabilizing output with monetary policy, part 1 (1 of 4)

- ▶ A negative supply shock shifts $SRAS$ up. *Without* intervention from the Central Bank, the economy would reach a new equilibrium at B .
- ▶ However, the Central Bank responds by increasing the money supply, shifting AD out, resulting in a new equilibrium at C (with higher price level, and no reduction in output).



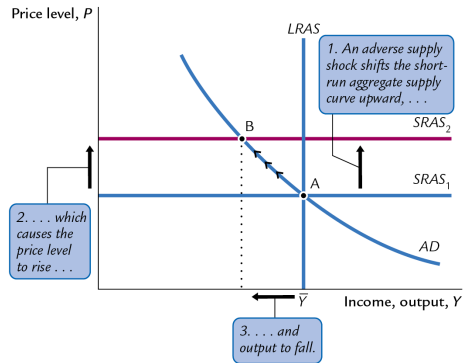
Stabilizing output with monetary policy, part 1 (2 of 4)

- Initial equilibrium (before the negative supply shock)



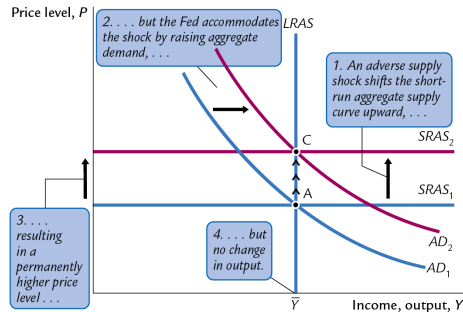
Stabilizing output with monetary policy, part 1 (3 of 4)

- ▶ A negative supply shock shifts $SRAS$ up. *Without* intervention from the Central Bank, the economy would reach a new equilibrium at B . Price level would be higher and output would be lower.
- ▶ However, the Central Bank will attempt to stabilize output.



Stabilizing output with monetary policy, part 1 (4 of 4)

- ▶ However, the Central Bank responds by increasing the money supply, shifting AD out, resulting in a new equilibrium at C (with higher price level and no reduction in output).
- ▶ The economy instead ends up at C , instead of B , due to the stabilization policy.



Analyzing the Covid-19 Recession

- ▶ Initially, the shock to the economy was an inward in LRAS.
 - ▶ Businesses closed
 - ▶ Businesses that remained open saw decreased productivity because of social distancing.
- ▶ However, as businesses closed, consumers lost the ability to spend money, AD shifted in due to C down.
 - ▶ Unable to dine-in at restaurants, travel, and attend concerts, movies, sporting events, museums, etc.

CHAPTER SUMMARY, PART 1

- ▶ Long run: Prices are flexible, output and employment are always at their natural rates, and the classical theory applies. Short run: Prices are sticky, and shocks can push output and employment away from their natural rates.
- ▶ Aggregate demand and supply: a framework to analyze economic fluctuations

CHAPTER SUMMARY, PART 2

- ▶ The aggregate demand curve slopes downward.
- ▶ The long-run aggregate supply curve is vertical because output depends on technology and factor supplies but not prices.
- ▶ The short-run aggregate supply curve is horizontal because prices are sticky at predetermined levels.

CHAPTER SUMMARY, PART 3

- ▶ Shocks to aggregate demand and supply cause fluctuations in GDP and employment in the short run.
- ▶ The Fed can attempt to stabilize the economy with monetary policy.

Appendix: Okun's law — a worked example

The empirical relation for the U.S. (Okun 1962, still roughly valid):

$$\% \Delta Y \approx 3\% - 2 \times \Delta u,$$

where Δu is the change in the unemployment rate (in percentage points).

Example 1 — forecast unemployment from output.

Real GDP grows 1% this year. Predict Δu :

$$1 = 3 - 2\Delta u \implies \Delta u = +1\text{pp.}$$

Below-trend growth (1% vs. 3% normal) raises unemployment by 1 percentage point.

Example 2 — forecast output from unemployment.

Unemployment jumps from 5% to 8% in a recession. Predict $\% \Delta Y$:

$$\% \Delta Y = 3 - 2(3) = -3\%.$$

GDP contracts about 3% that year.

◀ back

